

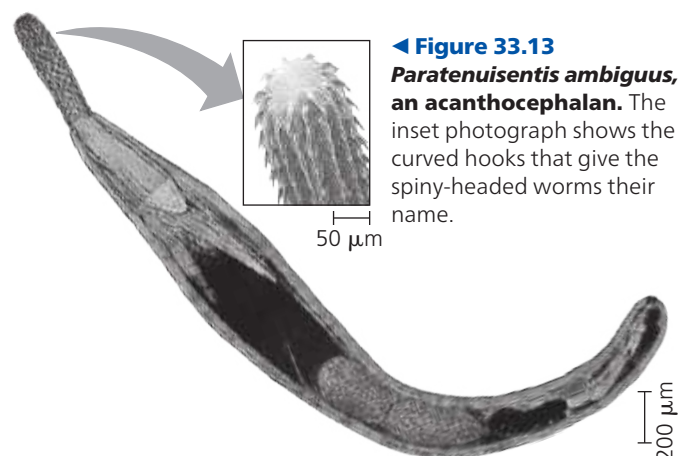
reproduction has certain advantages over asexual reproduction (see Concept 46.1). For example, species that reproduce asexually tend to accumulate harmful mutations in their genomes faster than sexually reproducing species. As a result, asexual species should experience higher rates of extinction.

Seeking to understand how they persist without males, researchers have been studying a clade of asexual rotifers named Bdelloidea. Some 360 species of bdelloid rotifers are known, and all of them reproduce by parthenogenesis without any males. Paleontologists have discovered bdelloid rotifers preserved in 35-million-year-old amber, and the morphology of these fossils resembles only the female form, with no evidence of males. Molecular clock analyses indicate that bdelloids have been asexual for over 50 million years. While it appears that they do not reproduce sexually, bdelloid rotifers may be able to generate genetic diversity in other ways. For example, bdelloids can tolerate very high levels of desiccation. When conditions improve and their cells rehydrate, DNA from other species enters their cells through cracks in the plasma membrane. Recent evidence suggests that this foreign DNA can be incorporated into the bdelloids' genome, thereby leading to increased genetic diversity.

Acanthocephalans

Acanthocephalans (1,100 species) are sexually reproducing parasites of vertebrates that lack a complete digestive tract and usually are less than 20 cm long. They are commonly called spiny-headed worms because of the curved hooks on the proboscis at the anterior end of their body (Figure 33.13). Acanthocephalans are triploblastic and, like rotifers, they have a hemocoel. Recent studies have shown that acanthocephalans originated from within the group traditionally known as Rotifera. In particular, rotifers in the genus *Seison* share a more recent common ancestor with acanthocephalans than they do with other rotifers, making the acanthocephalans a group of highly modified “rotifers.”

All acanthocephalans are parasites that have complex life cycles with two or more hosts. Some species manipulate the behavior of their intermediate hosts (generally arthropods)



◀ **Figure 33.13**
***Paratenuisentis ambiguus*,
an acanthocephalan.** The
inset photograph shows the
curved hooks that give the
spiny-headed worms their
name.

in ways that increase their chances of reaching their final hosts (generally vertebrates). For example, acanthocephalans that infect New Zealand mud crabs cause their hosts to move to more visible areas on the beach, where the crabs are more likely to be eaten by birds, the worms' final hosts.

Ectoprocts and Brachiopods

Bilaterians in the phyla Ectoprocta and Brachiopoda have a *lophophore*, a crown of ciliated tentacles around their mouth (see Figure 32.12a). As the cilia draw water toward the mouth, the tentacles trap suspended food particles. Other similarities, such as a U-shaped alimentary canal and the absence of a distinct head, reflect these organisms' sessile existence. In contrast to flatworms, which lack a body cavity, and rotifers, which have a hemocoel, ectoprocts and brachiopods have a coelom (see Figure 32.9a).

Ectoprocts (from the Greek *ecto*, outside, and *procta*, anus) are colonial animals that superficially resemble clumps of moss. (In fact, their common name, bryozoans, means “moss animals.”) In most species, the colony is encased in a hard exoskeleton studded with pores through which the lophophores extend (Figure 33.14a). Most ectoproct species live in the sea, where they are among the most widespread and numerous sessile animals. Several species are important reef builders. Ectoprocts also live in lakes and rivers. Colonies of the freshwater ectoproct *Pectinatella magnifica* grow on submerged sticks or rocks and can grow into a gelatinous, ball-shaped mass more than 10 cm across.

Brachiopods, or lamp shells, superficially resemble clams and other hinge-shelled molluscs, but the two halves of the brachiopod shell are dorsal and ventral rather than lateral, as in clams (Figure 33.14b). All brachiopods are marine. Most live attached to the seafloor by a stalk, opening their shell slightly to allow water to flow through the lophophore. The living brachiopods are remnants of a much richer past that included

▼ **Figure 33.14** Ectoprocts and brachiopods.



(a) Ectoprocts, such as this creeping bryozoan (*Plumatella repens*), are colonial.

(b) Brachiopods, such as this lampshell (*Terebratulina retusa*), have a hinged shell. The two parts of the shell are dorsal and ventral.